221TCS004	INTRODUCTION TO AI AND NLP	CATEGORY	L	T	P	CREDIT
		Program Core 2	3	0	0	3

Preamble:

This course introduces the concepts, tools, and techniques of machine learning for text data. The students will learn the elementary concepts as well as emerging trends in the field of NLP. This course helps the learners to extract information from unstructured text, identify linguistic structure of it, and to apply different the techniques for text analytics. The learners will able to implement and evaluate NLP applications using machine learning and deep learning methods.

Course Outcomes:

After the completion of the course, the student will be able to: *

CO1	Analyze the applications of AI in the domain of NLP **(Cognitive Knowledge Level: Apply)**
CO2	Transform text into an appropriate data structure **(Cognitive Knowledge Level: Apply)**
CO3	Apply Probability models, language models, and Markov models for Text processing **(Cognitive Knowledge Level: Apply)**
CO4	Build NLP applications using Machine Learning Methods **(Cognitive Knowledge Level: Apply)**
CO5	Design, Develop, Implement and Present innovative Ideas on NLP and AI **(Cognitive Knowledge Level: Create)**

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tools to model, analyze and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	✔		✔		✔	✔	
CO 2	✔		✔		✔	✔	
CO 3	✔		✔		✔	✔	
CO 4	✔	✔	✔	✔	✔	✔	
CO 5	✔	✔	✔	✔	✔	✔	✔

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	50-80%
Analyse	20-40%

Evaluate	Assess using Assignments/Project
Create	Assess using Assignments/Project

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

The evaluation shall only be based on application, analysis or design-based questions (for both internal and end-semester examinations).

Continuous Internal Evaluation: 40 marks
Micro project/Course based project : 20 marks
Course based task/Seminar/Quiz : 10 marks
Test paper, 1 no. : 10 marks

The project shall be done individually. Group projects are not permitted. Test paper shall include a minimum of 80% of the syllabus.

Course-based task/test paper questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students.

End Semester Examination Pattern:

The end-semester examination will be conducted by the University. There will be two parts; Part A and Part B. Part A contain 5 numerical questions with 1 question from each module, having 5 marks for each question. (Such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students). Students shall answer all questions.

Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions

Model Question Paper

QP CODE:

Reg No: _______________

Name: _________________ **PAGES : 4**

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: 221TCS004

Course Name: INTRODUCTION TO AI AND NLP

Max. Marks: 60 **Duration: 2.5 Hours**

PART A

Answer All Questions. Each Question Carries 5 Marks

1. a. Contrast NLTK and Spacy **5**

b. Explain the different types of artificial Intelligence

2. a. Analyse the importance of the concept of vectors in NLP **5**

b. Explain the simple functioning of the ELIZA chatbot

3. Explain Markov Property and Markov model for NLP **5**

4. Discuss topic modeling and the intuition behind using LDA for topic modeling **5**

5. Describe implementing binary text classification using TensorFlow **5**

(5x5=25)

Part B

(Answer any five questions. Each question carries 7 marks)

6. (a) Explain the applications of Artificial intelligence in Health care **(7)**

relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student should answer any five. Each question can carry 7 marks.

The total duration of the examination will be 150 minutes.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Implement artificial intelligence-based solution in agriculture for optimization of irrigation and application of pesticides and herbicides

2. Develop Artificial Intelligence/Machine Learning based for Diabetes Care

3. Explain the role of Natural Language Processing Applications in Finance

Course Outcome 2 (CO2)

1. Which of the following techniques can be used for keyword normalization in NLP, the process of converting a keyword into its base form?

a. Lemmatization

b. Soundex

c. Cosine Similarity

d. N-grams

Interpret your answer.

2. Explain any two out of the eight methods that **Convert Text to Features**

a. Recipe 1. One Hot encoding.

b. Recipe 2. Count vectorizer.

c. Recipe 3. N-grams.

d. Recipe 4. Co-occurrence matrix.

e. Recipe 5. Hash vectorizer.

f. Recipe 6. Term Frequency-Inverse Document Frequency (TF-IDF)

g. Recipe 7. Word embedding.

h. Recipe 8. Implementing fast Text.

3. Explain the following with an example

a. Regular expression and its working

b. Properties of regular expression

c. Meta characters Big brackets [] and search [abcd] in ‘kasdfaiabcasdfaabc’ and write the output

d. Let’s look at the following sentence: “I ate an apple and played the piano.” Generate the one-hot embedding matrix for this sentence

Course Outcome 3(CO3):

a. Implement TF-IDF from scratch
b. Explain how you will implement NLP in other languages
c. Explain Markov Model for text classifier and build a text classifier using the Markov model

Example: Take two sets of poems by two different authors, Edgar Allan Poe and Robert Frost. Given a sentence, the system should be able to predict the author

Course Outcome 4 (CO4):

1. Application: Latent Semantic Indexing for Search Engine Optimization using PCA/SVD

2. Implement ANN for multiclass classification

3. Implement Spam Detection using Naïve Bayes or Logistic regression

Course Outcome 5 (CO5):

1. Implement Text Summarization using python

2. How will you implement a sentiment analyzer in Python using logistic regression to predict sentiment on Amazon reviews?

3. Application: Topic Modeling Using Latent Dirichlet Allocation

7.	(a)	Elaborate vectors, their usefulness in NLP, tokenization, stop words, stemming, and lemmatization with examples	**(3)**
	(b)	Explain the limitations of the count vectorizer and the necessity of TF-IDF	**(2)**
	(c)	In a corpus of N documents, one randomly chosen document contains a total of T terms, and the term "hello" appears K times. What is the correct value for the product of TF (term frequency) and IDF (inverse-document-frequency), if the term "hello" appears in approximately one-third of the total documents?	**(2)**
8.	(a)	Explain the regular expression and need for a regular expression in NLP	**(3)**
	(b)	How do regular expressions work?	**(2)**
	(c)	Explain the use of common Regex functions used in NLP	**(2)**
9.	(a)	Explain the language model, applications of language modeling, method to Compute the probability of a sentence, and the curse of dimensionality	**(2)**
	(b)	what is Markov's assumption in language modeling and n-grams	**(3)**
	(c)	Implement n-grams and update-function	**(2)**
10.	(a)	Explain the spam detection problem and why you want to detect them (2)	**(3)**
	(b)	Apply the Markov model n-gram approach to solve this problem and implement in Python	**(4)**
11	(a)	Explain RNN for text classification in TensorFlow	**(2)**
	(b)	Explain Parts of Speech (PoS)Tagging in TensorFlow	**(3)**
	(c)	Explain Named Entity Recognition in TensorFlow	**(2)**
12.	(a)	Explain CNN for text classification	**(7)**

Syllabus

Module	Contents	Total Lecture Hours (40 hrs)
1	**INTRODUCTION TO ARTIFICIAL INTELLIGENCE:** Artificial Intelligence? History, AI on a conceptual level, Types of AI, Use Cases, importance and applications of AI, AI algorithms, types of machine learning, types of problems solved in AI, advantages, and disadvantages of AI, AI In Marketing, Banking, Finance, Agriculture, HealthCare, Gaming, Space Exploration., Autonomous Vehicles, Chatbots, Artificial Creativity, AI Tools & Frameworks, AI vs Machine Learning vs Deep Learning, an overview of python for AI, **INTRODUCTION TO NLP:** NLP in the Real World, NLP Tasks, Language? Its Building Blocks, Why Is NLP Challenging? Machine Learning, Deep Learning, and NLP: An Overview, Approaches to NLP, Heuristics-Based NLP, Machine Learning & Deep Learning for NLP, NLP Pipeline, Applications of NLP-Machine translation, Speech recognition, Image Captioning, spam detection, text prediction- Introduction to Software Packages-Spacy, NLTK, Gensim, PyTorch, **Regular Expression-** importance, properties, working and python package (re), case study: working of Eliza chatbot	7
2	**REGULAR EXPRESSION & TEXT PROCESSING:** Common regex function, Meta Characters- Big brackets, cap, Backslash, Squared Brackets, Special Sequences, Asterisk, Plus, And Question mark, Curly Brackets **Understanding Pattern Objects-** Match Method Vs Search Method, Finditer Method, Logical Or, Beginning And End Patterns, Parenthesis **String Modification**- split method, sub-method, subn method, **Text Processing**-Words, Tokens, Counting words, vocabulary, corpus, tokenization in spacy- **Sentiment Classification-** (yelp) download a review dataset use data preparation using NumPy, pandas, counter, re-add tokens to vocabulary, build vocabulary from a data frame, from corpus, one hot encoding, encoding documents, train test splits, feature computation, confusion matrix, analysis. **Language Independent Tokenization**: Types of tokenization — Word, Character, and sub-word tokenization, problems with word tokenizer, drawbacks of a character-based tokenizer, problems with sub-word tokenization, Byte Pair Encoding, , **String Matching and Spelling Correction**-Minimum edit distance- table filling, dynamic programming,	9
	WORD EMBEDDING & PROBABILISTIC MODELS: Vector Models & Text Preprocessing: Vectors, Bag of Words, Count Vectorizer, Tokenization, Stopwords, Stemming, and Lemmatization, Stemming, and Lemmatization, Count Vectorizer, Vector Similarity. TF-IDF, Word-to-Index Mapping, Building TF-IDF, Neural Word Embeddings, Neural Word Embeddings. Vector Models Text Pre-processing Summary, steps of NLP analysis, **Probabilistic Models-Language**	9

3	**Modelling:** importance, types of language modeling, the curse of dimensionality, Language Model Markov Assumption And N-Grams, Language Model Implementation – Setup, Ngrams Function, Update Counts Function, Probability Model Function, Reading Corpus, Language Model Implementation Sampling Text, **Markov Models:** Markov Property, Markov Model, Probability Smoothing and Log-Probabilities, Building a Text Classifier, **Article Spinning** – Problem, N-Gram Approach, implementation, **Cipher Decryption with Language Modeling And Genetic Algorithm** Ciphers, substitution cipher, bigrams, maximum likelihood, and log-likelihood, Language models, Genetic Algorithms,	
4	**NLP USING MACHINE LEARNING MODELS**-Spam Detection– Problem, Naive Bayes theorem, Intuition, spam detection using Naïve Bayes, class imbalance, ROC, AUC, AND F1 SCORE, Implementing spam detection in python, **Sentiment Analysis** -Problem, Logistic Regression Intuition, Multiclass Logistic Regression, Logistic Regression Training and Interpretation, sentiment analysis implementation in python, **Text Summarization**-Using Vectors, Text Rank Intuition,Text Rank in Python, Text Summarization in Python **Topic Modeling**-different topic modeling techniques, Latent Dirichlet Allocation (LDA) – Essentials, Latent Dirichlet Allocation–Topic Modeling with Latent Dirichlet, **Latent Symmatc Modelling(Indexing)**-LSA / LSI Introduction, Singular Value Decomposition Intuition, LSA / LSI: Applying SVD to NLP, Latent Semantic Analysis / Latent Semantic Indexing in Python	7
5	**DEEP LEARNING**- word embeddings, nonlinear neural networks, **Neuron** – Intro, Fitting a Line, Classification Code Preparation, Text Classification in Tensorflow, The Neuron, How does a model learn?, **Feed Forward Neural Networks**- Ann-introduction, The Geometrical Picture, Activation Functions, Multiclass Classification, Text Classification ANN in Tensorflow, Text Preprocessing Code Preparation, Text Preprocessing in Tensorflow, Embeddings, CBOW(continuous bag of words), CBOW in Tensorflow, **Convolution Neural Networks**- Convolution, pattern matching, weight sharing, convolution in color images, CNN Architecture, CNN for Text, CNN for NLP in Tensorflow, **Recurrent Neural Networks**- Simple RNN / Elman Unit, RNNs: Paying Attention to Shapes, GRU, and LSTM. RNN for Text Classification in TensorFlow, Parts-of-Speech Tagging, and Named Entity Recognition in TensorFlow	8

COURSE PLAN		
Sl.NO	**TOPIC**	**NO. OF LECTURES**
	Module 1 – Introduction of AI & NLP (7 hours)	
Module 1: Introduction To Artificial Intelligence (7 hours)		
1.1	Artificial Intelligence? History, AI on a conceptual level, Types of AI, Use Cases, importance and applications of AI,	**1**
1.2	AI algorithms, types of machine learning, types of problems solved in AI, advantages, and disadvantages of AI	**1**
1.3	AI In Marketing, Banking, Finance, Agriculture, HealthCare, Gaming, Space Exploration, Autonomous Vehicles, Chatbots,	**1**
1.4	Artificial Creativity, AI Tools & Frameworks, AI vs Machine Learning vs Deep Learning, an overview of python for AI,	**1**
INTRODUCTION TO NLP,		
1.5	NLP in the Real World, NLP Tasks, Language? Its Building Blocks, Why Is NLP Challenging? Machine Learning, Deep Learning, and NLP: An Overview, Approaches to NLP, Heuristics-Based NLP, Machine Learning & Deep Learning for NLP	**1**
1.6	NLP Pipeline, Applications of NLP-Machine translation, Speech recognition, Image Captioning, spam detection, text prediction-	**1**
1.7	Introduction to Software Packages-Spacy, NLTK, Gensim, PyTorch, **Regular Expression-** importance, properties, working and python package (re), case study: working of Eliza chatbot	**1**
Module 2- Regular Expression & Text Processing (9 Hours)		
2.1	Common regex function used in NLP, -	**1**
2.2	Meta Characters- Big brackets, cap, Backslash,	**1**
2.3	Squared Brackets, Special Sequences, Asterisk, Plus, And Question mark, Curly Brackets	**1**
2.4	**Understanding Pattern Objects-** Match Method Vs Search Method, Finditer Method, Logical Or, Beginning And End Patterns, Parenthesis	**1**
2.5	**String Modification**- split method, sub-method, subn method,	**1**
TEXT PROCESSING		
2.6	Words, Tokens, Counting words, vocabulary, corpus, tokenization in spacy-	**1**
2.7	**Sentiment Classification-** (yelp) download a review dataset use –data preparation using NumPy, pandas, counter, re-add tokens to vocabulary, build vocabulary from a data frame, from corpus, one hot encoding, encoding documents, train test splits, feature computation, confusion matrix, analysis.	**1**

2.8	**Language Independent Tokenization**: Types of tokenization — Word, Character, and sub-word tokenization, problems with word tokenizer, drawbacks of a character-based tokenizer, problems with sub-word tokenization, Byte Pair Encoding	1
2.9	**String Matching and Spelling Correction**-Minimum edit distance- table filling, dynamic programming,	1
Module 3-Word Embedding & Probabilistic Models (9 Hours)		
3.1	**Vector Models & Text Preprocessing:** Vectors, Bag of Words, Count Vectorizer, Tokenization, Stop words,	1
3.2	Stemming and Lemmatization, Count Vectorizer, Vector Similarity. TF-IDF,	1
3.3	Word-to-Index Mapping, Building TF-IDF	1
3.4	Neural Word Embeddings, Neural Word Embeddings Demo. Vector Models & Text Pre-processing Summary, steps of a typical NLP analysis	1
	PROBABLISTIC MODELS	
3.5	**Language Modelling:** types of language modeling, the importance of language modeling, the curse of dimensionality, Language Model Markov Assumption And N-Grams,	1
3.6	Language Model Implementation – Setup, Ngrams Function, Update Counts Function, Probability Model Function, Reading Corpus, Language Model Implementation Sampling Text,	1
3.7	**Markov Models:** Markov Property, Markov Model, Probability Smoothing and Log-Probabilities, Building a Text Classifier,	1
3.8	**Article Spinning**–Problem Description, N-Gram Approach, implementation in python,	1
3.9	**Cipher Decryption with Language Modeling And Genetic Algorithm-** Ciphers, substitution cipher, bigrams, maximum likelihood, and log-likelihood, Language models, Genetic Algorithms,	1
Module 4 – NLP Using Machine Learning Models (7 Hours)		
4.1	**Spam Detection–** Problem, Naive Bayes theorem, Intuition, spam detection using Naïve Bayes, class imbalance, ROC, AUC, AND F1 SCORE, Implementing spam detection in python,	1
4.2	**Sentiment Analysis** -Problem, Logistic Regression Intuition, Multiclass Logistic Regression,	1
4.3	Logistic Regression Training and Interpretation, sentiment analysis implementation in python	1
4.4	**Text Summarization**-Using Vectors, Text Rank Intuition	1
4.5	Text Rank in Python, Text Summarization in Python	1
4.6	**Topic Modeling**-different topic modeling techniques, Latent Dirichlet Allocation (LDA) – Essentials, Latent Dirichlet Allocation–Topic Modeling with Latent Dirichlet	1
4.7	**Latent Symmatc Modelling(Indexing)**-LSA / LSI Introduction, Singular Value Decomposition Intuition, LSA / LSI: Applying SVD to NLP, Latent Semantic Analysis / Latent Semantic Indexing in Python	1

Module 5 - Deep Learning (8 Hours)		
5.1	word embeddings, nonlinear neural networks	1
5.2	**Neuron** – Intro, Fitting a Line, Classification Code Preparation, Text Classification in Tensorflow, The Neuron, How does a model learn?,	1
5.3	**Feed Forward Neural Networks**- Ann-introduction, The Geometrical Picture, Activation Functions, Multiclass Classification, Text Classification ANN in Tensorflow,	1
5.4	Text Preprocessing Code Preparation, Text Preprocessing in Tensorflow, Embeddings, CBOW(continuous bag of words), CBOW in Tensorflow	1
5.5	CONVOLUTION NEURAL NETWORKS:- CNN-Introduction, Convolution, pattern matching, weight sharing, convolution in color images,	1
5.6	**Convolution Neural Networks**- Convolution, pattern matching, weight sharing, convolution in color images, CNN Architecture, CNN for Text, CNN for NLP in Tensorflow,	1
5.7	**Recurrent Neural Networks**- Simple RNN / Elman Unit, RNNs: Paying Attention to Shapes, , GRU, and LSTM. RNN for Text Classification in TensorFlow,	1
5.8	Parts-of-Speech Tagging, and Named Entity Recognition in TensorFlow	1

Reference Books

1. Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, Practical Natural Language Processing, Shroff/O'Reilly, 2020.

2. Steven Bird, Ewan Klein, Edward Loper, Natural Language Processing with Python, O'Reilly

3. Akshay Kulkarni, Adarsha Shivananda, Natural Language Processing Recipes Unlocking Text Data with Machine Learning and Deep Learning using Python, Apress

4. Taweh Beysolow II 'Applied Natural Language Processing with Python- Implementing Machine Learning and Deep Learning Algorithms for Natural Language Processing, Apress

5. Palash Goyal, Sumit Pandey, Karan Jain 'Deep Learning for Natural Language Processing Creating Neural Networks with Python, Apress

6. Hobson Lane. Cole Howard, Hannes Max Hapke 'Natural Language Processing in Action, Understanding, analyzing, and generating text with Python', ©2019 by Manning Publications Co

8. Wolfgang Ertel, Introduction to Artificial Intelligence, Springer,